Adjoint Equations of PDE constrained problem

Tags: #Numerical-Method #Differential_Equations #Examples #Variation

Introduction

We are going to illustrate some examples to show how to find adjoint equation of PDE constrained problem with calculus of variation.

1D PDE-constrained optimization problem

Let us consider an optimization problem which is given by

$$\min_m J(u,m) = \frac{1}{2}\int_0^1 (u-u_d)^2\,dx + \frac{\alpha}{2}\int_0^1 m^2\,dx$$

where the constraint is

$$-u''(x) = m(x), \quad u(0) = u(1) = 0$$

Show that the adjoint equation in this problem is

$$\lambda'' = u - u_d, \quad \lambda(0) = \lambda(1) = 0$$

Let us impose a functional

$$\phi = J(x,u,m) + \int_0^1 \lambda(x)(-u''-m)\,dx = \int_0^1 \mathcal{L}(x,u,m,\lambda)\,dx$$

where the Lagrangian $\mathcal{L}$ is defined as

$$\mathcal{L}(x,u,m,\lambda) = \frac{1}{2}(u-u_d)^2 + \frac{\alpha}{2}m^2 + \lambda(x)(-u''-m)$$

In order to apply the calculus of variation, we need to introduce some weak variations such that

$$\tilde{u} = u + \varepsilon\,\delta u, \quad \tilde{m} = m + \varepsilon\,\delta m$$

where

$$\delta u(0) = \delta u(1) = 0 \quad \text{and} \quad \delta m(0) = \delta m(1) = 0.$$

Hence,

$$\Delta\phi = \phi(\varepsilon) - \phi(0) = \int_1^0 (\mathcal{L}(x, u + \varepsilon\,\delta u, m + \varepsilon\,\delta m, \lambda) - \mathcal{L}(x, u, m, \lambda))\,dx$$

and the Taylor expansion of $\Delta\phi$ is

$$\Delta\phi = \delta\phi + \frac{1}{2}\delta^2\phi + \mathcal{O}(\varepsilon^3)$$

For now, we only consider the first variation to determine the condition of extremum, where

$$\delta\phi = \left.\frac{d\phi(\varepsilon)}{d\varepsilon}\right|_{\varepsilon=0} \varepsilon$$

and

$$\frac{d\phi(\varepsilon)}{d\varepsilon}|_{\varepsilon=0} = \int_0^1 ((u - u_d)\,\delta u + \alpha\, m\,\delta m - \lambda(x)\delta u'' - \lambda(x)\,\delta m)\,dx$$

Applying integral by parts, we obtain

$$\int_0^1 \lambda(x)\,\delta u''\,dx = \int_0^1 \lambda''(x)\,\delta u\,dx + [\lambda(x)\,\delta u']_0^1 - [\lambda'(x)\,\delta u]_0^1$$

Thus,

$$\frac{d\phi(\varepsilon)}{d\varepsilon}|_{\varepsilon=0} = \int_0^1 ((u - u_d - \lambda'')\,\delta u + \alpha\, m\,\delta m - \lambda(x)\,\delta m)\,dx + [\lambda(x)\,\delta u']_0^1$$

To reach the extremum, we should have

$$\lambda''(x) = u(x) - u_d$$

where $\lambda(x)$ can be chosen as $\lambda(0) = \lambda(1) = 0$.

Adjoint Boundary Control Method for Bi-harmonic Type Problem

Let us consider a system of PDEs

$$\begin{cases} \nabla^2\psi = \omega \\ \nabla^2\omega = f(\psi, \omega) \end{cases}$$

with Dirichlet and Neumann boundary conditions of ψ. This problem can be converted to an optimization problem:

$$\min_\eta J(\tilde{\psi}, \eta) = \min_\eta \frac{1}{2}\int_{\partial\Omega} (\frac{\partial\tilde{\psi}}{\partial n}(\eta) - \mathbf{u}_\Gamma \cdot \mathbf{t})^2\,ds + \frac{\alpha}{2}\int_{\partial\Omega} \eta^2\,ds$$

with the PDE constraints

$$r(\tilde{\psi}, \omega, \eta) = \nabla^2\tilde{\psi} - \omega = 0$$
$$s(\tilde{\psi}, \omega, \eta) = \nabla^2\omega - f(\tilde{\psi}, \omega) = 0$$

where η is called the *control variable*, and it is the boundary value of ω

$$\eta = \omega|_{\partial\Omega}$$

$\mathbf{u}_\Gamma \cdot \mathbf{t} = A(x)$ is a prescribed boundary condition of ψ and α is a real number.

1. Determine the adjoint equations for two-dimensional steady state Stokes problem.
2. Determine the adjoint equations for two-dimensional steady state Navier-Stokes Equation.

Related note:

- [[Note-Continuous and Discrete Adjoints to the Euler Equations for Fluids]]
- [[Adjoint Boundary Control for a Bi-harmonic type of System-Calculus of Variation]]

1. The adjoint equations for two-dimensional Stokes problem

For the Stokes problem, the constraint $s(\omega, \eta)$ is

$$s(\omega, \eta) = \nabla^2 \omega(\eta) = 0 \quad \forall x \in \Omega$$

Let us consider a Lagrangian functional such that

$$\mathcal{L}(\tilde{\psi}, \omega, \eta, p, q) = \frac{1}{2} \int_{\partial\Omega} (\frac{\partial \tilde{\psi}}{\partial n} - \mathbf{u}_\Gamma \cdot \mathbf{t})^2 \, ds + \frac{\alpha}{2} \int_{\partial\Omega} \eta^2 \, ds + \int_\Omega p(\nabla^2 \tilde{\psi} - \omega) \, d\Omega + \int_\Omega q(\nabla^2 \omega) \, d\Omega$$

We define weak variations such that

$$\begin{aligned} \tilde{\psi}^* &= \tilde{\psi} + \varepsilon \, \delta\tilde{\psi} \\ \omega^* &= \omega + \varepsilon \, \delta\omega \\ \eta^* &= \eta + \varepsilon \, \delta\eta \end{aligned}$$

where

$$\begin{aligned} \delta\tilde{\psi}|_{\partial\Omega} &= 0 \\ \delta\omega|_{\partial\Omega} &= \delta u \end{aligned}$$

For notational convenience, we can consider the Lagrangian functional $\mathcal{L}(\tilde{\psi}, \omega, \eta, p, q)$ as a function of ε, thus

$$\mathcal{L}(\varepsilon) = \mathcal{L}(\tilde{\psi}^*, \omega^*, \eta^*, p, q)$$

and the total variation is

$$\Delta\mathcal{L} = \mathcal{L}(\varepsilon) - \mathcal{L}(0) = \delta\mathcal{L} + \frac{1}{2}\delta^2\mathcal{L} + O(\varepsilon^3)$$

The first variation of $\mathcal{L}$ is defined as

$$\delta\mathcal{L} = \frac{d\mathcal{L}}{d\varepsilon}\bigg|_{\varepsilon=0} \varepsilon$$

where

$$\begin{aligned}\frac{d\mathcal{L}}{d\varepsilon} =& \frac{dJ(\tilde{\psi}+\varepsilon\psi)}{d\varepsilon} \\ &+ \frac{d}{d\varepsilon}\int_\Omega p(\nabla^2(\tilde{\psi}+\varepsilon\,\delta\tilde{\psi}) - (\omega+\varepsilon\,\delta\omega))\,d\Omega \\ &+ \frac{d}{d\varepsilon}\int_\Omega q(\nabla^2(\omega+\varepsilon\,\delta\omega))\,d\Omega\end{aligned}$$

Since

$$\begin{aligned}J(\tilde{\psi}+\varepsilon\,\tilde{\psi},\eta+\varepsilon\,\delta\eta) = \frac{1}{2}\int_{\partial\Omega}(\frac{\partial\tilde{\psi}}{\partial n}+\varepsilon\frac{\partial\,\delta\tilde{\psi}}{\partial n})^2\,ds - \int_{\partial\Omega}(\frac{\partial\tilde{\psi}}{\partial n}+\varepsilon\frac{\partial\,\delta\tilde{\psi}}{\partial n})(\mathbf{u}_\Gamma\cdot\mathbf{t})\,ds + \frac{1}{2}\int_{\partial\Omega}(\mathbf{u}_\Gamma\cdot\mathbf{t})^2\,ds \\ + \frac{\alpha}{2}\int_{\partial\Omega}(\eta^2+2\varepsilon\eta\,\delta\eta+\varepsilon^2(\delta\eta)^2)\,ds\end{aligned}$$

Thus,

$$\frac{dJ(\tilde{\psi}+\varepsilon\psi,\eta+\varepsilon\,\delta\eta)}{d\varepsilon} = \int_{\partial\Omega}(\frac{\partial\tilde{\psi}}{\partial n}-\mathbf{u}_\Gamma\cdot\mathbf{t})\frac{\partial\,\delta\tilde{\psi}}{\partial n}\,ds + \int_{\partial\Omega}\varepsilon\frac{\partial\,\delta\tilde{\psi}}{\partial n}\,ds + \int_{\partial\Omega}\alpha\,\eta\,\delta\eta\,ds + \varepsilon\int_{\partial\Omega}\alpha\,(\delta\eta)^2\,ds$$

and therefore

$$\left.\frac{dJ}{d\varepsilon}\right|_{\varepsilon=0} = \int_{\partial\Omega}(\frac{\partial\tilde{\psi}}{\partial n}-\mathbf{u}_\Gamma\cdot\mathbf{t})\frac{\partial\,\delta\tilde{\psi}}{\partial n}\,ds + \int_{\partial\Omega}\alpha\,\eta\,\delta\eta\,ds$$

Let

$$\begin{aligned}R(\tilde{\psi}+\varepsilon\,\delta\tilde{\psi},\omega+\varepsilon\,\delta\tilde{\omega}) &= \int_\Omega p(\nabla^2(\tilde{\psi}+\varepsilon\,\delta\tilde{\psi}) - (\omega+\varepsilon\,\delta\omega))\,d\Omega \\ &= \int_\Omega p(\nabla^2\tilde{\psi}-\omega)\,d\Omega + \varepsilon\int_\Omega p(\nabla^2\,\delta\tilde{\psi}-\delta\omega)\,d\Omega\end{aligned}$$

then

$$\left.\frac{dR(\tilde{\psi}+\varepsilon\,\delta\tilde{\psi},\omega+\varepsilon\,\delta\omega)}{d\varepsilon}\right|_{\varepsilon=0} = \int_\Omega p(\nabla^2\,\delta\tilde{\psi}-\delta\omega)\,d\Omega$$

By the second Green's Identity

$$\int_\Omega (f\,\nabla^2 g - g\,\nabla^2 f)\,d\Omega = \int_{\partial\Omega}(f\frac{\partial g}{\partial n} - g\frac{\partial f}{\partial n})ds$$

we obtain

$$\int_\Omega (p\nabla^2\,\delta\tilde{\psi} - \delta\tilde{\psi}\,\nabla^2 p)\,d\Omega = \int_{\partial\Omega}(p\frac{\partial\,\delta\tilde{\psi}}{\partial n} - \delta\tilde{\psi}\frac{\partial p}{\partial n})\,ds = \int_{\partial\Omega} p\frac{\partial\,\delta\tilde{\psi}}{\partial n}\,ds$$

because $\delta\tilde{\psi}=0$ on the boundary. Thus

$$\left.\frac{dR(\tilde{\psi}+\varepsilon\,\delta\tilde{\psi},\omega+\varepsilon\,\delta\omega)}{d\varepsilon}\right|_{\varepsilon=0} = \int_\Omega \nabla^2 p\,\delta\tilde{\psi}\,d\Omega + \int_{\partial\Omega} p\frac{\partial\,\delta\tilde{\psi}}{\partial n}\,ds - \int_\Omega p\,\delta\omega\,d\Omega$$

Similarly, let

$$S(\omega + \varepsilon\,\delta\omega) = \int_\Omega q\nabla^2(\omega + \varepsilon\,\delta\omega)\,d\Omega$$

and we have

$$\frac{dS(\omega + \varepsilon\,\delta\omega)}{d\varepsilon}\bigg|_{\varepsilon=0} = \int_\Omega q\nabla^2\,\delta\omega\,d\Omega$$

By the second Green's Identity, it yields

$$\int_\Omega q\nabla^2\,\delta\omega\,d\Omega = \int_\Omega \delta\omega\,\nabla^2 q\,d\Omega + \int_{\partial\Omega} q\frac{\partial\,\delta\omega}{\partial n}\,ds - \int_{\partial\Omega} \delta\omega\frac{\partial q}{\partial n}\,ds$$

Combining these expressions, it renders us the following expression

$$\begin{aligned}\frac{d\mathcal{L}}{d\varepsilon}\bigg|_{\varepsilon=0} &= \int_{\partial\Omega}(\frac{\partial\tilde{\psi}}{\partial n} - \mathbf{u}_\Gamma\cdot\mathbf{t} + p)\frac{\partial\,\delta\tilde{\psi}}{\partial n}\,ds + \int_{\partial\Omega}\alpha\,\eta\,\delta\eta\,ds + \int_\Omega(\nabla^2 q - p)\,\delta\omega\,d\Omega\\ &+ \int_{\partial\Omega} q\frac{\partial\,\delta\omega}{\partial n}\,ds - \int_{\partial\Omega}\delta\omega\frac{\partial q}{\partial n}\,ds + \int_\Omega \nabla^2 p\,\delta\tilde{\psi}\,d\Omega\end{aligned}$$

The weak variation of ω on the boundary implies that

$$\int_{\partial\Omega}\eta\,\delta\eta\,ds = \int_{\partial\Omega}\eta\,\delta\omega\,ds$$

On the other hand, since the variation of $\mathcal{L}$ can be expressed with the *gradients of functional* such that

$$\begin{aligned}\delta\mathcal{L} &= \nabla_{\tilde{\psi}}\mathcal{L}\cdot\delta\tilde{\psi} + \nabla_\omega\mathcal{L}\cdot\delta\omega + \nabla_\eta\mathcal{L}\cdot\delta\eta\\ &= \int_{\partial\Omega}\nabla_{\tilde{\psi}}\mathcal{L}\,\delta\tilde{\psi}\,ds + \int_\Omega\nabla_\omega\mathcal{L}\,\delta\omega\,d\Omega + \int_{\partial\Omega}\nabla_\eta\mathcal{L}\,\delta\eta\,ds\end{aligned}$$

where

$$\int_{\partial\Omega}\nabla_\eta\mathcal{L}\,\delta\eta\,ds = \int_{\partial\Omega}(\alpha\eta - \frac{\partial q}{\partial n})\delta\eta\,ds$$

The necessary condition for the functional $\mathcal{L}$ for an extremum implies that

$$\delta\mathcal{L} = \frac{d\mathcal{L}}{d\varepsilon}\bigg|_{\varepsilon=0}\varepsilon = 0$$

Thus the Euler-Lagrange's equations are

$$\begin{aligned}&\nabla^2 p = 0 \quad \forall x\in\Omega\\ &\nabla^2 q = p \quad \forall x\in\Omega\\ &p = -(\frac{\partial\tilde{\psi}}{\partial n} - \mathbf{u}_\Gamma\cdot\mathbf{t}) \quad \forall x\in\partial\Omega\\ &q = 0 \quad \forall x\in\partial\Omega\\ &\alpha\,\eta - \frac{\partial q}{\partial n} = 0 \quad \forall x\in\partial\Omega\end{aligned}$$

The last equation is the *optimality condition*, the others are *adjoint equations*.

Remarks

- Each argument of the Lagrangian functional is considered to be independent of the others, but it is no longer true for the objective functional because the system state and control variable are constrained.
- The gradient of a functional is different from the gradient of real value function of multi-variables, because it is defined for infinite-dimensional space.

2. The adjoint equations for two-dimensional steady state Navier-Stokes flow

2.1 The general case

We are going to illustrate how to determine the adjoint equations for the stream function-vorticity formulation of a steady state Navier-Stokes equation.

The stream function-vorticity formulation is given by

$$\begin{aligned}
&\nabla^2 \tilde{\psi} = \omega \\
&\nabla^2 \omega - Re(u \cdot \nabla)\omega = 0 \\
&u = -\nabla \times \tilde{\psi}\mathbf{e}_z
\end{aligned}$$

where u is the velocity field of the flow.

The objective functional is

$$J(\tilde{\psi}, \eta) = \frac{1}{2}\int_{\partial\Omega} (\frac{\partial \tilde{\psi}}{\partial n} - a)^2 \, ds + \frac{\alpha}{2}\int_{\partial\Omega} \eta^2 \, ds$$

where η represents the boundary vorticity and it is our control variable. And the prescribed boundary Neumann condition of ψ is given by

$$\frac{\partial \psi}{\partial n} = u_\Gamma \cdot \mathbf{t} = a(x)$$

We introduce a Lagrangian functional such that

$$\begin{aligned}
\mathcal{L}(\tilde{\psi}, \omega, u, \eta, \beta, \gamma, \lambda) &= J(\tilde{\psi}, \eta) + R(\tilde{\psi}, \omega, \beta) + S(\omega, u, \gamma) + T(\tilde{\psi}, u, \lambda) \\
&= \frac{1}{2}\int_{\partial\Omega} (\frac{\partial \tilde{\psi}}{\partial n} - a)^2 \, ds + \frac{\alpha}{2}\int_{\partial\Omega} \eta^2 \, ds \\
&+ \int_\Omega \beta(\nabla^2 \tilde{\psi} - \omega) \, d\Omega + \int_\Omega \gamma(\nabla^2 \omega - Re(u \cdot \nabla)\omega) \, d\Omega \\
&+ \int_\Omega \lambda \cdot (\nabla \times \tilde{\psi}\mathbf{e}_z + u) \, d\Omega
\end{aligned}$$

Let us set the weak variations

$$\tilde{\psi}^* = \tilde{\psi} + \varepsilon\, \delta\tilde{\psi}$$
$$\omega^* = \omega + \varepsilon\delta\omega$$
$$\eta^* = \eta + \varepsilon\delta\eta$$
$$u^* = u + \varepsilon\delta u$$

where

$$\delta\tilde{\psi}|_{\partial\Omega} = 0$$
$$\delta\omega|_{\partial\Omega} = \delta\eta$$

and we can consider the Lagrangian functional as a function of ε, meaning

$$\begin{aligned}\mathcal{L}(\varepsilon) =& J(\tilde{\psi}^*, \eta^*) + R(\tilde{\psi}^*, \omega^*, \beta) + S(\omega^*, u^*, \gamma) + T(\tilde{\psi}^*, u^*, \lambda)\\ =& \frac{1}{2}\int_{\partial\Omega} (\frac{\partial\tilde{\psi}^*}{\partial n} - a)^2\, ds + \frac{\alpha}{2}\int_{\partial\Omega} (\eta^*)^2\, ds\\ &+ \int_\Omega \beta(\nabla^2\tilde{\psi}^* - \omega^*)\, d\Omega + \int_\Omega \gamma(\nabla^2\omega^* - Re(u^*\cdot\nabla)\omega^*)\, d\Omega\\ &+ \int_\Omega \lambda\cdot(\nabla\times\tilde{\psi}^*\mathbf{e}_z + u^*)\, d\Omega\end{aligned}$$

Therefore

$$\left.\frac{dJ(\varepsilon)}{d\varepsilon}\right|_{\varepsilon=0} = \int_{\partial\Omega} (\frac{\partial\tilde{\psi}}{\partial n} - a)\frac{\partial\,\delta\tilde{\psi}}{\partial n}\, ds + \int_{\partial\Omega} \alpha\eta\,\delta\eta\, ds$$

and

$$\left.\frac{dR(\varepsilon)}{d\varepsilon}\right|_{\varepsilon=0} = \int_\Omega \nabla^2\beta\,\delta\tilde{\psi}\, d\Omega + \int_{\partial\Omega} \beta\frac{\partial\delta\tilde{\psi}}{\partial n}\, ds - \int_\Omega \beta\,\delta\omega\, d\Omega$$

Since

$$\nabla\cdot u = 0$$

thus we have

$$\nabla\cdot(\omega\, u) = (u\cdot\nabla)\omega$$

and we can write the vorticity equation in conservative form

$$\nabla^2\omega - Re\nabla\cdot(\omega\, u) = 0$$

and

$$S(\varepsilon) = \int_\Omega \gamma\{\nabla^2(\omega + \varepsilon\delta\omega) - Re(\nabla\cdot(u\,\omega) + \varepsilon\nabla\cdot(u\,\delta\omega + \omega\,\delta u) + \varepsilon^2\nabla\cdot(\delta\omega\,\delta u))\}\, d\Omega$$

Therefore

$$\left.\frac{dS(\varepsilon)}{d\varepsilon}\right|_{\varepsilon=0} = \int_\Omega \gamma\nabla^2\delta\omega\, d\Omega - \{Re\int_\Omega \gamma(\nabla\cdot(u\,\delta\omega) + \nabla\cdot(\delta u\,\omega))\, d\Omega\}$$

By the divergence theorem, we obtain

$$\begin{aligned}\int_\Omega \gamma\nabla\cdot(u\,\delta\omega)\,d\Omega &= \int_\Omega \nabla\cdot(\gamma\,\delta\omega\,u)\,d\Omega - \int_\Omega u\,\delta\omega\cdot\nabla\gamma\,d\Omega \\ &= \int_{\partial\Omega} \gamma\delta\omega\,u\cdot n\,ds - \int_\Omega u\,\delta\omega\cdot\nabla\gamma\,d\Omega\end{aligned}$$

and

$$\begin{aligned}\int_\Omega \gamma\nabla\cdot(\delta u\,\omega)\,d\Omega &= \int_\Omega \nabla\cdot(\gamma\,\omega\,\delta u)\,d\Omega - \int_\Omega \omega\,\delta u\cdot\nabla\gamma\,d\Omega \\ &= \int_{\partial\Omega} \gamma\omega\,\delta u\cdot n\,ds - \int_\Omega \omega\,\delta u\cdot\nabla\gamma\,d\Omega\end{aligned}$$

The previous expressions imply that

$$\begin{aligned}\left.\frac{dS(\varepsilon)}{d\varepsilon}\right|_{\varepsilon=0} &= \int_\Omega \gamma\nabla^2\delta\omega\,d\Omega + Re\int_\Omega u\,\delta\omega\cdot\nabla\gamma\,d\Omega + Re\int_\Omega \omega\,\delta u\cdot\nabla\gamma\,d\Omega \\ &\quad - Re\int_{\partial\Omega} \gamma\delta\omega\,u\cdot n\,ds - Re\int_{\partial\Omega}\gamma\omega\,\delta u\cdot n\,ds\end{aligned}$$

For the constraint of velocity, we have

$$T(\varepsilon) = \int_\Omega \lambda\cdot(\nabla\times(\tilde{\psi}+\varepsilon\,\delta\tilde{\psi})\mathbf{e}_z + (u+\varepsilon\,\delta u))\,d\Omega$$

where

$$\nabla\times(\tilde{\psi}+\varepsilon\,\delta\tilde{\psi})\mathbf{e}_z = \nabla\times\tilde{\psi}\mathbf{e}_z + \varepsilon\nabla\times\delta\tilde{\psi}\mathbf{e}_z$$

Thus

$$\left.\frac{dT(\varepsilon)}{d\varepsilon}\right|_{\varepsilon=0} = \int_\Omega \lambda\cdot(\nabla\times\delta\tilde{\psi}\mathbf{e}_z + \delta u)\,d\Omega$$

Rearranging those terms, we have the expression of the gradient of Lagrangian functional

$$\begin{aligned}\left.\frac{d\mathcal{L}}{d\varepsilon}\right|_{\varepsilon=0} &= \int_{\partial\Omega}\left(\frac{\partial\tilde{\psi}}{\partial n}-a\right)\frac{\partial\,\delta\tilde{\psi}}{\partial n}\,ds + \int_{\partial\Omega}\alpha\eta\,\delta\eta\,ds \\ &\quad + \int_\Omega \nabla^2\beta\,\delta\tilde{\psi}\,d\Omega + \int_{\partial\Omega}\beta\frac{\partial\delta\tilde{\psi}}{\partial n}\,ds - \int_\Omega \beta\,\delta\omega\,d\Omega \\ &\quad + \int_\Omega \gamma\nabla^2\delta\omega\,d\Omega + Re\int_\Omega u\,\delta\omega\cdot\nabla\gamma\,d\Omega + Re\int_\Omega \omega\,\delta u\cdot\nabla\gamma\,d\Omega \\ &\quad - Re\int_{\partial\Omega}\gamma\delta\omega\,u\cdot n\,ds - Re\int_{\partial\Omega}\gamma\omega\,\delta u\cdot n\,ds \\ &\quad + \int_\Omega \lambda\cdot(\nabla\times\delta\tilde{\psi}\mathbf{e}_z+\delta u)\,d\Omega\end{aligned}$$

In addition, if we apply the curl formula

$$\int_\Omega (\nabla\times u)\cdot v\,d\Omega = \int_\Omega u\cdot(\nabla\times v)\,d\Omega + \int_{\partial\Omega}(n\times u)\cdot v\,dS$$

and the convention

$$\mathbf{t} \times \mathbf{n} = -\mathbf{e}_z$$

where $\mathbf{t}$ is the tangent vector of the boundary, we will obtain

$$\begin{aligned}
\int_\Omega \lambda \cdot (\nabla \times \delta\tilde{\psi}\mathbf{e}_z)\, d\Omega &= \int_\Omega \delta\tilde{\psi}\mathbf{e}_z \cdot (\nabla \times \lambda)\, d\Omega + \int_{\partial\Omega} (n \times \delta\tilde{\psi}\mathbf{e}_z) \cdot \lambda\, ds \\
&= \int_\Omega \delta\tilde{\psi}\mathbf{e}_z \cdot (\nabla \times \lambda)\, d\Omega - \int_{\partial\Omega} \lambda \cdot \mathbf{t}\, \delta\tilde{\psi}\, ds
\end{aligned}$$

Thus the gradient of Lagrangian functional is

$$\begin{aligned}
\left.\frac{d\mathcal{L}}{d\varepsilon}\right|_{\varepsilon=0} &= \int_{\partial\Omega} (\frac{\partial\tilde{\psi}}{\partial n} - a)\frac{\partial\, \delta\tilde{\psi}}{\partial n}\, ds + \int_{\partial\Omega} \alpha\eta\, \delta\eta\, ds \\
&+ \int_\Omega \nabla^2\beta\, \delta\tilde{\psi}\, d\Omega + \int_{\partial\Omega} \beta\frac{\partial\delta\tilde{\psi}}{\partial n}\, ds - \int_\Omega \beta\, \delta\omega\, d\Omega \\
&+ \int_\Omega \gamma\nabla^2\delta\omega\, d\Omega + Re\int_\Omega u\, \delta\omega \cdot \nabla\gamma\, d\Omega + Re\int_\Omega \omega\, \delta u \cdot \nabla\gamma\, d\Omega \\
&- Re\int_{\partial\Omega} \gamma\delta\omega\, u \cdot n\, ds - Re\int_{\partial\Omega} \gamma\omega\, \delta u \cdot n\, ds \\
&+ \int_\Omega \delta\tilde{\psi}\mathbf{e}_z \cdot (\nabla \times \lambda)\, d\Omega - \int_{\partial\Omega} \lambda \cdot \mathbf{t}\, \delta\tilde{\psi}\, ds + \int_\Omega \lambda \cdot \delta u\, d\Omega
\end{aligned}$$

Substituting the no-penetration condition into the functional gradient, it yields

$$\begin{aligned}
\left.\frac{d\mathcal{L}}{d\varepsilon}\right|_{\varepsilon=0} &= \int_{\partial\Omega} (\frac{\partial\tilde{\psi}}{\partial n} - a + \beta)\frac{\partial\, \delta\tilde{\psi}}{\partial n}\, ds \\
&+ \int_\Omega (\nabla^2\beta + (\nabla \times \lambda) \cdot \mathbf{e}_z)\delta\tilde{\psi}\, d\Omega \\
&+ \int_{\partial\Omega} (\alpha\eta - \frac{\partial\gamma}{\partial n})\delta\eta\, ds \\
&+ \int_\Omega (\nabla^2\gamma + Re(u \cdot \nabla\gamma) - \beta)\, \delta\omega\, d\Omega \\
&+ \int_{\partial\Omega} \gamma\frac{\partial\, \delta\omega}{\partial n}\, ds + \int_\Omega (Re(\omega\nabla\gamma) + \lambda) \cdot \delta u\, d\Omega - \int_{\partial\Omega} \lambda \cdot \mathbf{t}\, \delta\tilde{\psi}\, ds
\end{aligned}$$

which implies the adjoint equations

$$\begin{aligned}
&\frac{\partial\tilde{\psi}}{\partial n} - a + \beta = 0 \quad \text{on } \partial\Omega \\
&\nabla^2\beta + (\nabla \times \lambda) \cdot \mathbf{e}_z = 0 \quad \text{in } \Omega \\
&\nabla^2\gamma + Re(u \cdot \nabla\gamma) - \beta = 0 \quad \text{in } \Omega \\
&\gamma = 0 \quad \text{on } \partial\Omega \\
&Re(\omega\nabla\gamma) + \lambda = 0 \quad \text{in } \Omega
\end{aligned}$$

and the gradient of the objective functional is

$$\nabla_\eta J = \alpha\eta - \frac{\partial \gamma}{\partial n}$$

We also have the sensitivity equations

$$\begin{aligned}
&\nabla^2 \delta\omega - Re\nabla \cdot (u\,\delta\omega + \omega\,\delta u) = 0 \quad \text{in } \Omega \\
&\nabla^2 \delta\tilde{\psi} - \delta\omega = 0 \quad \text{in } \Omega \\
&\nabla \times \delta\tilde{\psi}\mathbf{e}_z + \delta u = 0 \quad \text{in } \Omega
\end{aligned}$$

2.2 The case with frozen velocity

The formulation with frozen velocity can be written as

$$\begin{aligned}
&\nabla^2 \tilde{\psi} = \omega \\
&\nabla^2 \omega - Re(\nabla \cdot (u_0 \omega)) = 0
\end{aligned}$$

The adjoint equations of this case are reduced to

$$\begin{aligned}
\frac{\partial \tilde{\psi}}{\partial n} - \mathbf{u}_\Gamma \cdot \mathbf{t} + \beta = 0 &\quad \text{on } \partial\Omega \\
\nabla^2 \beta = 0 &\quad \text{in } \Omega \\
\nabla^2 \gamma + Re(u_0 \cdot \nabla\gamma) - \beta = 0 &\quad \text{in } \Omega \\
\gamma = 0 &\quad \text{on } \partial\Omega
\end{aligned}$$

and the gradient of the objective functional is

$$\nabla_\eta J = \alpha\eta - \frac{\partial \gamma}{\partial n}$$

Thought

The adjoint equations can be derived from variational calculus on a Lagrangian, however there are some limitations for a moderate or even high Reynolds number. For the Stokes flow, the adjoint equations are not coupled; but for the Navier-Stokes equation, the adjoint equations are coupled, and we have to deduce the gradient of objective functional with those coupled PDE.

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